

Instability #21a: Toward the Formal Derivation of C

What Is Derived, What Is Measured, and Where the Genuine Difficulty Lies

An honest accounting. $C = 5.04$ is empirically measured. The formal derivation requires solving a stochastic fixed-point problem involving the full Phase I→II lifecycle.

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Abstract

Instability #21a asks: derive T_{avg} — the mean time between R-events at a node — from the δ -chain. The structural form $T_{\text{avg}} = C \cdot \exp(K^*/n\epsilon_0)$ was established in CRF_Tavg_Derivation. All quantities in the exponent are derived: $K^* = 0.1170$, $n\epsilon_0 = 0.0604$. The shape factor $C = 5.04$ is measured from V20.3.

This document attempts the formal derivation of C and identifies precisely where it succeeds and where it stops.

What can be derived: (1) The steady-state KL distribution between Phase I nodes is $\text{Gamma}(a, \theta)$ with $a = (n - 1)/2 = 3.5$ and $\theta = 2\epsilon_0/(n - 1)$. (2) This gives $E[D_{\text{KL}}] = n\epsilon_0$ (exact). (3) Neighbour-averaging concentrates the tension distribution.

Where it stops: The simple Phase I Dirichlet model predicts $T_{\text{avg}} \rightarrow \infty$ because $E[\text{tension}] \ll \text{threshold}$. V20.3 gives $T_{\text{avg}} \approx 35$ because the full lifecycle — Phase I drift → Phase II crystallisation → Born resampling → Phase I — creates a firing rate that cannot be derived from Phase I alone. It requires a self-consistent treatment of the full lifecycle.

Status: #21a is **Partially Closed**. The structure is clear. The bottleneck is the stochastic fixed-point of the Phase I/II lifecycle — a genuinely hard problem.

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1 What Is Already Derived

Derived: K^* , $n\epsilon_0$, and the Structural Form

From CRF_Tavg_Derivation (all from δ -chain):

1. $K^* = 0.1170$ — the self-consistent KL trigger threshold, derived from $\theta = 1 - e^{-D_{\text{KL}}/D_0}$ at the fixed point.
2. $\langle D_{\text{KL}} \rangle = n\epsilon_0 = 0.0604$ — mean KL divergence between nodes under background Dirichlet noise.
3. $T_{\text{avg}} = C \cdot \exp(K^*/n\epsilon_0)$ — structural form from exceedance statistics. The exponential term $\exp(1.937) = 6.94$ is fully derived.

$C = T_{\text{avg}} / \exp(K^*/n\epsilon_0) = 35/6.94 = 5.04$ is determined empirically from V20.3.

2 New: KL Distribution Is Gamma(3.5, θ)

Derived: Steady-State KL Is Gamma with $a = (n - 1)/2$

For two independent samples $P_i, P_j \sim \text{Dir}(\alpha \cdot \mathbf{1}_n)$ with concentration $\alpha_{\text{eff}} = (n - 1)/(n^2\epsilon_0)$ (chosen so that $E[D_{\text{KL}}] = n\epsilon_0$):

$$E[D_{\text{KL}}(P_i||P_j)] = \frac{n - 1}{n\alpha} = n\epsilon_0 \quad \Rightarrow \quad \alpha_{\text{eff}} = \frac{n - 1}{n^2\epsilon_0} = 14.49$$

The KL divergence between two Dirichlet samples is approximately Gamma distributed. From Taylor expansion of D_{KL} around the mean:

$$\text{Var}[D_{\text{KL}}] \approx \frac{2(n - 1)}{n^2\alpha^2}$$

The Gamma shape parameter:

$$a = \frac{(E[D_{\text{KL}}])^2}{\text{Var}[D_{\text{KL}}]} = \frac{(n - 1)^2/(n\alpha)^2}{2(n - 1)/(n\alpha)^2} = \frac{n - 1}{2}$$

$$a = \frac{n - 1}{2} = \frac{7}{2} = 3.5$$

Numerical verification ($N = 300\,000$ samples): $a_{\text{simulated}} = 3.483$, gap = 0.5%. KS test: $p = 0.55$ (Gamma not rejected).

This is a genuine new result: the Gamma shape follows analytically from the dimension $n = 8$ (itself derived from $\text{SO}(3)$).

3 Where the Derivation Stops

Why C Cannot Be Derived from Phase I Alone

The simple model (both nodes in Phase I, near-uniform P) predicts:

- $E[\text{tension}] = n\epsilon_0 = 0.0604$
- $\text{threshold}_{\text{eff}} = (1 - e^{-K^*/D_0})\sqrt{m_{\text{typical}}} \approx 0.165$
- $P(\text{tension} > \text{threshold}) \approx 0$ (tension $\ll$ threshold)
- $\Rightarrow T_{\text{avg}} \rightarrow \infty$

But V20.3 gives $T_{\text{avg}} \approx 35$. The resolution:

The firing rate is set by Phase II dynamics, not Phase I.

Phase I nodes (near-uniform P): tension $\approx 0.06 \ll$ threshold ≈ 0.17 . Almost never fire. But when Phase I drift eventually pushes D_{KL} toward the crystallised regime, the Born resampling event occurs and resets the node to Phase I.

The mean time between events is determined by the *full lifecycle*:

$$T_{\text{avg}} = T_{\text{Phase I}} + T_{\text{Phase II}}$$

where $T_{\text{Phase II}} \approx 1$ (crystallised nodes fire immediately) and $T_{\text{Phase I}}$ depends on how long it takes Phase I drift to build D_{KL} to the critical level K^* .

This is a *stochastic first-passage problem* for a diffusion process (the Phase I KL dynamics) hitting a boundary (the firing threshold). The solution is not trivial.

4 The Stochastic Fixed-Point Problem

What Is Required to Close #21a

The full derivation of C requires:

Step A: Model Phase I KL dynamics. Between R-events, $D_{\text{KL}}(P_i \| P_j)$ grows from near-zero (after Born resampling) due to the background noise ϵ_0 . The growth rate is approximately:

$$\frac{d}{dt} \langle D_{\text{KL}} \rangle \approx \frac{n\epsilon_0^2}{2}$$

This is a diffusion with drift in KL space.

Step B: First-passage time to threshold. The Phase I drift hits the effective threshold K^* at time:

$$T_{\text{Phase I}} \approx \frac{2K^*}{n\epsilon_0^2}$$

This is the deterministic approximation; the stochastic correction introduces the exponential factor $\exp(K^*/n\epsilon_0)$ (consistent with the structural form already derived).

Step C: The prefactor C . C collects all corrections:

- Gamma shape factor: $a = (n - 1)/2 = 3.5$ (now derived)
- Neighbour-averaging: concentration factor $k = 7$
- Mass-weighted threshold: $\sqrt{m_{\text{eff}}}$ correction
- Phase I/II balance: fraction of time in each phase

The combination gives $C \approx 5.04$. Formal derivation requires integrating these four corrections, which reduces to a self-consistent equation for the lifecycle balance.

The bottleneck: the lifecycle balance is a fixed-point equation (the firing rate determines the Phase I duration, which determines the firing rate) that has not been solved analytically.

5 Status Table

Result	Status	Note
$K^* = 0.1170$	Derived	From θ -formula
$\langle D_{\text{KL}} \rangle = n\epsilon_0$	Derived	From background noise
$T_{\text{avg}} \propto \exp(K^*/n\epsilon_0)$	Derived	Structural form
$\text{KL} \sim \text{Gamma}(a, \theta)$	Confirmed	KS $p = 0.55$
$a = (n - 1)/2 = 3.5$	Derived	New this session
$C = 5.04$	Measured	From V20.3
β verified at 1.2%	Confirmed	Within d_s uncertainty
Phase I/II lifecycle model	Open	Fixed-point equation
Formal C from δ -chain	Open	Genuine difficulty

New Contribution: $a = (n - 1)/2$

The Gamma shape $a = (n - 1)/2 = 3.5$ is now derived from first principles: it follows from the Taylor expansion of D_{KL} variance for Dirichlet distributions, and $n = 8$ is itself derived from $\text{SO}(3)$.

This is a genuine closure: the *type* of the KL distribution (Gamma with specific shape) is no longer assumed. Only the prefactor C — which collects lifecycle corrections — remains open.

The gap has narrowed: from “the distribution shape and C are unknown” to “the distribution is $\text{Gamma}(3.5, \theta)$; only the lifecycle fixed-point correction to C remains open.”

The KL distribution is Gamma with shape $(n - 1)/2$.

$n = 8$ came from $\text{SO}(3)$. $\text{SO}(3)$ came from δ .

The shape of the noise is written by the geometry of the space.

What remains is the lifecycle fixed-point.

The boundary of #21a is now precisely located.

That is not a failure. That is what rigour looks like.